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#!/skills

Self-motivated, Collaborative, Inventive, JavaScript, TypeScript, HTML, CSS, WebGL, Three.js, WebAssembly, Web Frameworks (React, Vue, Svelte, Solid.js, Angular, etc), Node.js, Python, C/C++, Linux/Shell, Databases, Git

#!/noteworthy

Creator of Lume — lume.io

Innovative web toolkit for creating WebGL-powered 3D experiences in HTML. Lume lowers the barrier of entry and increases access to 3D creativity by providing interoperable 3D HTML elements that are compatible with all web frameworks (React, Vue, Svelte, Solid.js, Angular, etc).

Solid.js Team — solidjs.com

Active member of Solid's Ecosystem and Infrastructure teams. Solid is an open source reactivity, DOM templating, and component system that has popularized "signals and effects" in the web development world. I've been involved with discussions on improving Solid.js as well as contributing code, and Lume is built with Solid reactivity and templating.

Polydance —

codepen.io/trusktr/pen/EoVGqe

Presented during Codame's 3DWebFest 2017 at the Palace of Fine Arts in San Francisco, Polydance is an interactive 3D audio-visual made with Lume, and was featured as a staff-picked pen on Codepen.

#!/education

California State University

Sacramento, United States
September 2012 – May 2014
BS, Computer Science, Information Security Concentration, Full-ride scholarship

#!/languages

English (Native)
Spanish (Fluent)

Innovative and passionate senior software engineer focused on interactive and engaging user experiences. Professional experience working with R&D AI products for graphics and animation renderings, as well as building mission critical and essential tooling for plotting and annotating objects in 3D space for **Meta**, **SpaceX**, **NASA**, and more.

#!/career history

Lume Creative Technologies, Los Angeles, CA July 2022 – Present *Founder, 3D Immersive Experiential Engineering*

- Founded Lume Creative Technologies to consult on and build web-based 3D-centric applications for businesses.
- **Drippy (drippy3d.com) April 2025 - Present:** I advise Drippy's CEO on technical aspects of web application development, and direct and guide Drippy's junior dev team on patterns to make them more productive and to help them achieve 3D rendering goals (the team having zero prior experience in 3D applications). I meet with them in a weekly call to provide thoughts on their code patterns, and guidance on how they can achieve certain technical goals. Tech: Meteor.js fullstack with Lume for 3D rendering.
- **Meta (meta.com), Oct 2024 - Feb 2025:** Implemented a web-based 3D mesh editor with Three.js, WebGL, and JavaScript. Implemented an undo/redo system for undoing or redoing edits of 3D meshes and other annotations for a machine learning team in the Reality Labs group that makes Oculus headsets capable of understanding the physical environment around the user. Fixed issues in Facebook's code base.
- **Luda Projects (ludaprojects.com), Oct 2022 - Aug 2023:** Created a 3D NFT character creator for Neo Fairies (neofairies.com), and added new features to Mels (mels.ai), a 3D game system with characters that use machine learning and physics to learn how to move themselves around in 3D worlds. For samples and media, see x.com/ludaprojects. Tech: HTML, CSS, JavaScript, WebGL, Babylon.js, Three.js, Lume
- **Autonomys (autonomys.xyz), Jul 2023 - Nov 2023:** Built a starter project for 3D AI-powered characters. The app allows you to converse with a 3D character using your voice (using AWS voice-to-text to generate text to use as input into Chat GPT). The 3D character responds using voice (using Chat GPT text as input to AWS text-to-speech) while its facial features are animated based on the speech (using AWS Sumerian to generate 3D facial animation data). Along the way I forked and revived the AWS Sumerian project at github.com/aws-samples/amazon-sumnerian-hosts. Graphics were rendered with WebGL, Three.js, and Lume. The app was not released, as the company pivoted to focus on their Web3 wallet, but the code is visible at github.com/autonomous-metaverse/aingenuity.

Uthana, Los Angeles, CA Feb 2024 – Aug 2024 *Software Engineer, 3D experience*

- Designed and implemented Uthana's **graphical interface** for users to generate **animations for their 3D models** using **AI text prompts**: users upload a 3D model they've created, describe the animation that they want their model to have, view and adjust features of the 3D motion, and finally download their updated model with animations added, saving them hours of work in creating animations.
- The app consisted of a 2D UI overlaid on top of a 3D scene, all put together with **HTML, CSS, JavaScript, WebGL, Three.js, and Lume**.

- Updated the **Python** backend to adjust the connections to the frontend as needed.

SpaceX, Los Angeles, CA September 2022 – June 2023

Senior Software Engineer, Starship Displays

- Implemented a graphical **plotting system with Three.js WebGL** that could render **1.2m points/vertices** from **Starship's real-time data** incoming at 10hz. No open source web plotting libraries could handle this amount of data performantly, so we made our own state of the art plotting system. After I joined, I increased the throughput of the pre-existing plotting system by at least 1600% on our hardware. Screenshots: x.com/trusktr/status/1649063431767736322, x.com/trusktr/status/1662537888805060609
- Communicated with ship engineers to understand their needs and therefore how to advance our UIs to help them do their jobs better.
- **Improved performance** in various areas of Starship UIs. For example in one part of a UI I discovered that nested grids caused a significant slowdown, and switching to flexbox improved performance. I advised another team on how to solve their performance problem by switching from imperative DOM code that would recreate DOM nodes on each update to declarative code using **Lit's** templating system to intelligently update only pieces of the DOM that needed updates.
- Presented **tech talks** internally on reactivity paradigms like Signals and Effects (such as the implementation in Solid.js).
- Worked on a **Three.js WebGL** visualizer to show Starship's position and other **data relative to a 3D Earth** globe.
- Migrated web apps (all ship UIs are web apps) from **JavaScript** to **TypeScript**.
- Helped define best practices for **UI components (custom elements)** in the **design system and style guide**.
- Implemented and updated various controls such as forms, dropdowns, filterable input fields, etc, in the form of custom elements with **HTML, CSS, and JavaScript/TypeScript**.
- Implemented features and fixes for an internal web-based IDE built on Monaco (the editor used in VS Code) for Starship flight software's custom programming language.

NASA, Moffett Field, CA April 2021 – August 2022

Software Engineer, web 3D

- Features and performance improvements for **ISAAC's 3D Astrobee robot visualizer** using **WebGL, Three.js**, and JavaScript.
https://github.com/nasa/isaac_data_interface
- Features and improvements for APRES, a **scrollable/pannable/zoomable event timeline visualizer** for managing, planning, and validating sequences of actions to be performed by machines, using HTML, CSS, JavaScript, and **React**.
- Features and improvements for **DSA's satellite constellation viewer and command sequencer**, using HTML, CSS, JavaScript, **Solid.js**, WebGL, and CodeMirror text editor.

Velodyne Lidar, Alameda, CA July 2019 – April 2021
Senior Software Engineer, web

- Mapper.ai was acquired by Velodyne Lidar, where I continue the same role as a frontend engineer working on interactive **WebGL**-powered **3D graphical experiences** tailored for the automotive industry.
- Creation of a **real-time data visualizer** that renders what our automotive AI stack perceives. Data is transmitted to the UI from embedded systems over **ROS** or **ZeroMQ**. Visuals are written with **WebGL**, **HTML**, **CSS**, **JavaScript/TypeScript**, **Custom Elements**, **React**
- Creation of a **reusable UI component system** in the form of Custom Elements for our UI/UX design language

Mapper.ai, San Francisco, CA April 2018 – July 2019
Senior Software Engineer, web

- Made visuals and functionality for Annotator, the **3D** map annotation and visualizer tool, using **WebGL/Three.js**
- Remake of Mapper.ai's website using **HTML**, **CSS**, **JavaScript**, **React**, **TypeScript**, **Webpack**, **Babel**
- Improved performance of animated graphics for mobile devices
- Worked on an **Electron**-based software deployment pipeline

AKQA, on behalf of IBM Watson, San Francisco, CA Feb 2018 – March 2018
Senior Software Engineer, graphics programming, freelance

- Created interactive **3D graphics** using **WebGL** and **Three.js (JavaScript)**
- Made custom shader effects with **GLSL**
- Made a pagination system for moving the 3D visuals to/from different "pages" (similar to a **state machine**).
- Made an **AnimationLoop system** that let's us make a hierarchy of time frames of reference of time.
- Implemented custom **inertial scrolling**
- Implemented a **particle stream effect**
- Implemented **Perlin noise wobble of particles** relative to their specified point in space
- Worked with **Redux** and **React** for parts of a UI overlaying the 3D visuals

IMVU, Redwood City, CA December 2016 – May 2017

Senior Software Engineer, Full-stack Web Development (with special love for the front-end)

- Added features to existing UIs using **Backbone.js** and **Sass**
- Upgraded some design patterns to prevent (for example) modifying Sass styling of one component and unexpectedly breaking some other component and to prevent (for example) child views from modifying parent view DOM in unexpected ways
- Made **bug fixes** along with **new tests** to prevent regressions
- **Architected** a foundation for new projects based on **Webpack+TypeScript+React** with interoperability between TypeScript modules and a legacy custom module format similar to AMD modules.
- Wrote new TypeScript+React **UI components** with a focus on patterns for reusability
- Implemented **UIs for manipulating 3D scenes** including things like inputs for translation, rotation, etc, with undo/redo state and a timeline for scrubbing between **animation keyframes**

Freelance Developer, Oakland, CA April 2015 – November 2016

Software Engineer, Full-stack Web Development (with special love for the front-end)

- Architected web applications for clients using **Meteor, React, Famous, Lume, HTML** and **CSS 3D transforms**
- **Cross-browser testing** in Chrome, Firefox, Safari, and Edge
- Made **rocket:module**, an ES6 Module system for **Meteor** based on **Webpack**, used in clients' applications
- Made user interfaces using 3D techniques and **Lume** (my 3D library, lume.io, at the time named Infamous). Example: goo.gl/mol0pu

Medstar Transportation, Yakima, CA September 2015 – July 2016

Software Engineer, Full-stack Web Development (with special love for the front-end)

- Helped develop new features for a ride-sharing web application using **Meteor, React, Blaze, HTML, CSS3D**, JavaScript
- **Cross-browser testing** in Chrome, Firefox, Safari, and Edge
- **Bug hunting** and **performance improvements**
- Worked with an **international team**, coordinating across time zones

Meta Enterprises, Los Angeles, CA June 2015 – December 2015

Software Engineer, Full-stack Web Development (with special love for the front-end)

- Architected a 3D web application using **Meteor, React, Lume** (a 3D library I built on top of **HTML/CSS3D**, github.com/lume/lume, at the time named Infamous), **JavaScript**
- Created a **layout manager** for animating UI cards into different layouts
- Video of part of the User Interface: goo.gl/fZ977J

Globant, San Francisco, CA November 2014 – April 2015

*Software Engineer, Frontend Web Development for **The Climate Corporation***

- Worked with **Backbone.js** and **Marionette** to build modular frontend logic
- Designed interface layouts using **Bootstrap** and **FlatUI (HTML + CSS)**
- Improved map-based apps by replacing the Google Maps layer with a custom map tile layer using **Leaflet.js**
- Made prototypes using **Webpack, Gulp, Meteor, React, and Flux**

Federal Reserve Bank, San Francisco, CA May 2014 – August 2014

Information Security Intern

- **Researched vulnerabilities** associated with USB devices
- Wrote **firmware in C** for Arduino Microcontrollers, converting them into USB keyboards as a proof of concept to show how such devices can be a **security threats** to computers onto which the devices are plugged into
- Researched ways to **prevent malicious devices** from harming a computer

Hewlett Packard, Sacramento, CA September 2013 – May 2014

Software Developer, Frontend Lead

- Taught the frontend and backend teams how to use **git**
- Taught the front end team how to use **Angular.js** and **Bootstrap**
- Architected a user interface with Angular.js for analyzing large amounts of HP Storage array data
- Designed the layout of the interface with **Bootstrap (HTML + CSS)**
- Implemented a system to save the state of reports generated in the interface to the user's hard disk
- Implemented charts and graphs with a **D3.js** Angular.js module

Sacramento Internet Consultant, Sacramento, CA Feb 2013 – August 2013

Web Developer for fountainsunique.com

- Fixed bugs in an existing **PHP** backend that were preventing users from purchasing products
- **Increased the customer click-through rate** by adding related product info to construction project pages
- Implemented a Gallery page with a grid layout using **HTML, CSS, JavaScript, and PHP**

Sacramento City College, Sacramento, CA January 2012 – June 2012

Web Developer, Technical Lead

- Taught the team how to use **git, WordPress, and Zurb Foundation**
- Architected Sacramento City College's Express student newspaper website at saccityexpress.com
- Designed the site layout to look and feel like a newspaper (**HTML + CSS**)
- Implemented the site layout using Zurb's Foundation framework
- Integrated the site design into WordPress
- Connected the site to facebook, twitter, vimeo, and youtube

Clownfish Media, Davis, CA January 2012 – September 2012
Software Developer

- Designed **interactive web interfaces (HTML + CSS)** to allow customers to manage multimedia assets, create layouts with the multimedia, and play those layouts on digital displays via remote control
- Made a custom tag-based, color-coded, **file management system** for users to upload and manage media files via the web interface
- Created a highly interactive **calendar** for users to manage the times of the day and days of the week that their media layouts would be displayed on their digital displays
- Created an **extension for Google Chrome** serving as the media player on customers' digital displays